

A proof of Conjecture 00000000027

$r_3(N) = N \exp(-O(\sqrt{\log N}))$ (Behrend's bound)

1 The conjecture and its reading

Conjecture 00000000027 reads: $r_3(N)$ (the maximal size of a subset of $[N]$ with no three-term arithmetic progression) satisfies $r_3(N) = N \cdot \exp(-O(\sqrt{\log N}))$. The Chinese text states the same formula.

Reading. With the usual meaning of the O -notation inside a formula, the statement asserts that there is a function g with $g(N) = O(\sqrt{\log N})$ as $N \rightarrow \infty$ and $r_3(N) = Ne^{-g(N)}$ for all large N . Because $1 \leq r_3(N) \leq N$ for $N \geq 1$, this is the same as the existence of a constant C with $Ne^{-C\sqrt{\log N}} \leq r_3(N)$ for all large N , i.e. Behrend's lower bound. We prove it with the explicit constant $C = 4$, for every $N \geq 1$, and with $0 \leq g(N) \leq 4\sqrt{\log N}$ (so g is non-negative and the O -bound holds with constant 4).

What is not claimed. The one-sided notation $-O(\cdot)$ does not assert a matching upper bound, and we do *not* prove one: the sharpness statement $r_3(N) \leq Ne^{-c\sqrt{\log N}}$ for some $c > 0$ (a Θ -version) is not claimed here. For context, the Wikipedia article on Salem–Spencer sets (retrieved; quoted in the attempt file) records upper bounds of the much weaker shape $\exp(-c(\log N)^{1/12})N$ (Kelley and Meka) with the exponent later improved to $1/9$ (Bloom and Sisask).

Conventions. $\log$ is the natural logarithm and $\sqrt{\cdot}$ the non-negative square root. $[N] = \{1, \dots, N\}$. A three-term arithmetic progression is $a, a + d, a + 2d$ with common difference $d \geq 1$ (trivial progressions a, a, a are excluded, as usual). We also show that the value of $r_3(N)$ is the same for the convention $[N] = \{0, \dots, N - 1\}$.

2 Definitions (as in the Lean file)

Definition 1 (C27.HasThreeAP, C27.r3). A finite set $A \subseteq \mathbb{N}$ has a three-term AP if there are $a, d \in \mathbb{N}$ with $d \geq 1$ and $a, a + d, a + 2d \in A$. Then

$$r_3(N) = \max\{|A| : A \subseteq \{1, \dots, N\}, A \text{ has no three-term AP}\},$$

formalized as the supremum of `card` over the filtered powerset of `!cc 1 N` (the empty set qualifies, so this is a genuine maximum).

Mathlib's `ThreeAPFree s` means: for all $a, b, c \in s$, $a + c = b + b$ implies $a = b$. Mathlib's `addRothNumber s` is the largest size of a `ThreeAPFree` subset of the finset s , and `rothNumberNat N = addRothNumber (range N)`, the $\{0, \dots, N - 1\}$ version.

3 Proof

Lemma 2 (`not_hasThreeAP_iff`). *For a finset $A \subseteq \mathbb{N}$: A has no three-term AP iff `ThreeAPFree` A .*

Proof. If $a, a + d, a + 2d \in A$ with $d \geq 1$, then $a + (a + 2d) = (a + d) + (a + d)$ but $a \neq a + d$. Conversely, let $a, b, c \in A$ with $a + c = 2b$ and $a \neq b$. If $a < b$, put $d = b - a$; then $b = a + d$ and $c = a + 2d$. If $a > b$, then $c < b$; put $d = b - c$, so $b = c + d$ and $a = c + 2d$. $\square$

Lemma 3 (`r3_eq_addRothNumber`, `r3_eq_rothNumberNat`). $r_3(N) = \text{addRothNumber}(\{1, \dots, N\}) = \text{rothNumberNat}(N)$.

Proof. The first equality follows from the previous lemma by comparing the two maxima in both directions. For the second, $\{1, \dots, N\} = [1, N + 1)$ and Mathlib's `addRothNumber_Ico` gives `addRothNumber` $([a, b)) = \text{rothNumberNat}(b - a)$ (translation invariance). $\square$

Lemma 4 (`r3_le`, `behrend_r3`, `r3_two_sided`). *For every N : $Ne^{-4\sqrt{\log N}} \leq r_3(N) \leq N$.*

Proof. The upper bound is Mathlib's `rothNumberNat_le`. The lower bound is Mathlib's `Behrend.roth_lower_bound : (N : R) * exp (-4 * sqrt (log N)) <= rothNumberNat N`, a formalization (by Y. Dillies and B. Mehta, file `Mathlib/Combinatorics/Additive/AP/Three/Behrend.lean`) of Behrend's construction [Behrend 1946], transported by the previous lemma. $\square$

Theorem 5 (`C27.conjecture_27`). *Let $g(N) = \log(N/r_3(N))$. Then $g = O(\sqrt{\log N})$ as $N \rightarrow \infty$; for every $N \geq 1$, $0 \leq g(N) \leq 4\sqrt{\log N}$; and for every $N \geq 1$, $r_3(N) = Ne^{-g(N)}$.*

Proof. For $N \geq 1$ the previous lemma gives $r_3(N) \geq Ne^{-4\sqrt{\log N}} > 0$, so $g(N)$ is the logarithm of a positive number and $Ne^{-g(N)} = N \cdot r_3(N)/N = r_3(N)$ (`r3_explicit`). From $r_3(N) \leq N$ we get $N/r_3(N) \geq 1$, so $g(N) \geq 0$; from $N \leq r_3(N)e^{4\sqrt{\log N}}$ we get $g(N) \leq 4\sqrt{\log N}$. Hence $|g(N)| \leq 4\sqrt{\log N}$ for all $N \geq 1$, which is the `IsBigO` statement `g =O[atTop] fun N => sqrt (log N)`. $\square$

4 Lean formalization

File `lean/Conjecture27/Basic.lean` (137 lines, only `import Mathlib`), namespace `C27`. Main theorem:

```
theorem conjecture_27 :
  exists g : Nat -> Real, (g =O[atTop] fun N : Nat => sqrt (log N)) /\
    (forall N, 1 <= N -> 0 <= g N /\ g N <= 4 * sqrt (log N)) /\
    forall N, 1 <= N -> (r3 N : Real) = N * exp (-g N)
```

(written here in ASCII; the Lean source uses the usual Unicode symbols). Further theorems: `not_hasThreeAP_iff`, `r3_eq_addRothNumber`, `r3_eq_rothNumberNat`, `r3_le`, `behrend_r3`, `r3_pos`, `r3_explicit`, `r3_two_sided`.

Axiom audit. `lake build` succeeds, and `#print axioms` for `C27.conjecture_27`, `C27.r3_two_sided` and `C27.r3_eq_rothNumberNat` reports only `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

Attribution and sources

The result is Behrend’s theorem; the contribution here is the faithful statement of the conjecture’s objects and its derivation from Mathlib.

- F. A. Behrend, *On Sets of Integers Which Contain No Three Terms in Arithmetical Progression*, Proc. Natl. Acad. Sci. USA (1946), <https://doi.org/10.1073/pnas.32.12.331> (bibliographic record retrieved from Crossref).
- Wikipedia, *Salem–Spencer set*, https://en.wikipedia.org/wiki/Salem%E2%80%93Spencer_set: “The construction of Salem and Spencer was improved by Felix Behrend in 1946, who found sets of size $n/e^{O(\sqrt{\log n})}$.”
- Mathlib, `Mathlib/Combinatorics/Additive/AP/Three/Behrend.lean` (`Behrend.roth_lower_bound`) and `Mathlib/Combinatorics/Additive/AP/Three/Defs.lean` (`rothNumberNat`, `addRothNumber_Ico`).